

Python Magic (Dunder) Methods Assignment

Instructions: Implement each required magic method in the specified class. Write clean Python code, create appropriate objects to test your implementation, and display the output to verify that each method works correctly.

Question 1: `__init__`, `__str__`, and `__setitem__`

Create a `Student` class with attributes `name`, `roll_no`, and `marks`. Implement `__init__` to initialize values, `__str__` to print the object in a readable format, and `__setitem__` to update attributes using dictionary-style syntax such as `student['marks']=90`. Demonstrate each method.

Question 2: `__add__`

Create a `Vector` class representing a 2D vector (`x`, `y`). Implement `__add__` so two vectors can be added using the `+` operator. Print the resulting vector.

Question 3: `__eq__`

Implement `__eq__` in the `Student` class so two students are equal if they have the same roll number. Test the comparison with different objects.

Question 4: `__len__` and `__getitem__`

Create a `Playlist` class that stores a list of songs. Implement `__len__` to return the total number of songs and `__getitem__` to access songs using indexing.

Question 5: `__call__`

Create a `Multiplier` class that accepts a number in the constructor. Implement `__call__` so the object behaves like a function and returns the multiplication result.